

Progressive Semantic Segmentation

MagNet: A Multi-Scale Segmentation Framework for High-Resolution Images

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Presented in CVPR 2021

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Key Focus:

High Resolution Images
Memory Efficiency (GPU
Constraints)

Core Strategy:

Multi-Scale Processing
Coarse-to-Fine- Refinement

Downsampling and Patch Processing

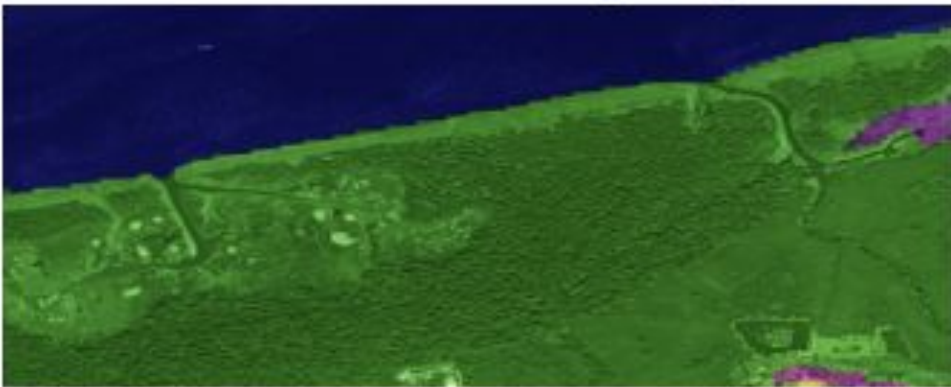


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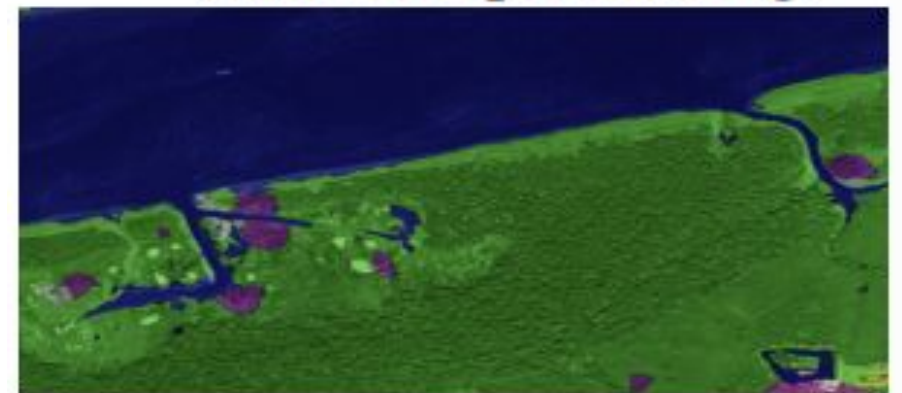
(a) Input image



(c) Downsampling



(d) Patch processing



GLNet (Global-Local Network) ↘

2021

Topology:

Parallel Dual-Branch Architecture

Input Processing:

Independently

Fusion Strategy:

One-Shot Aggregation

MagNet ↗

Topology:

Serial Multi-Stage

Input Processing:

Cumulative

Fusion Strategy:

Progressive Refinement

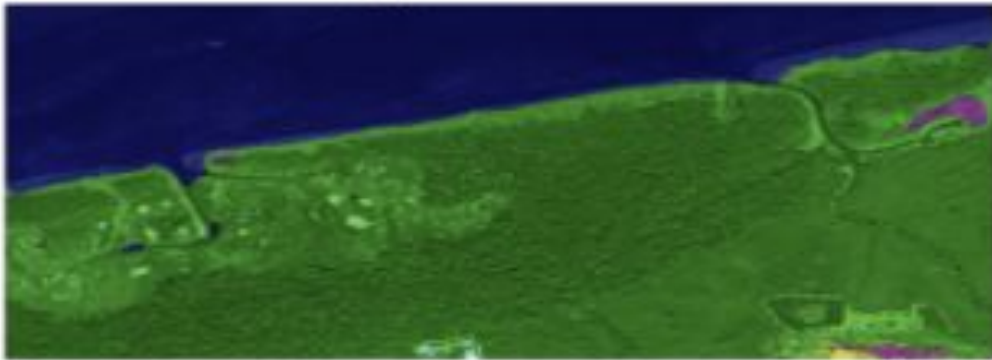
Key Innovation:

Uncertainty-Gated Module

GLNet vs MagNet

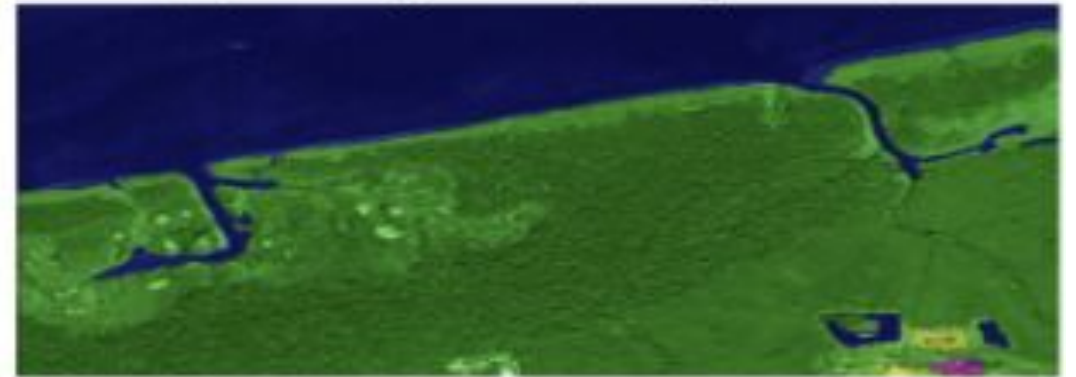
GLNet (Global-Local Network) 2021

(e) GLNet



MagNet

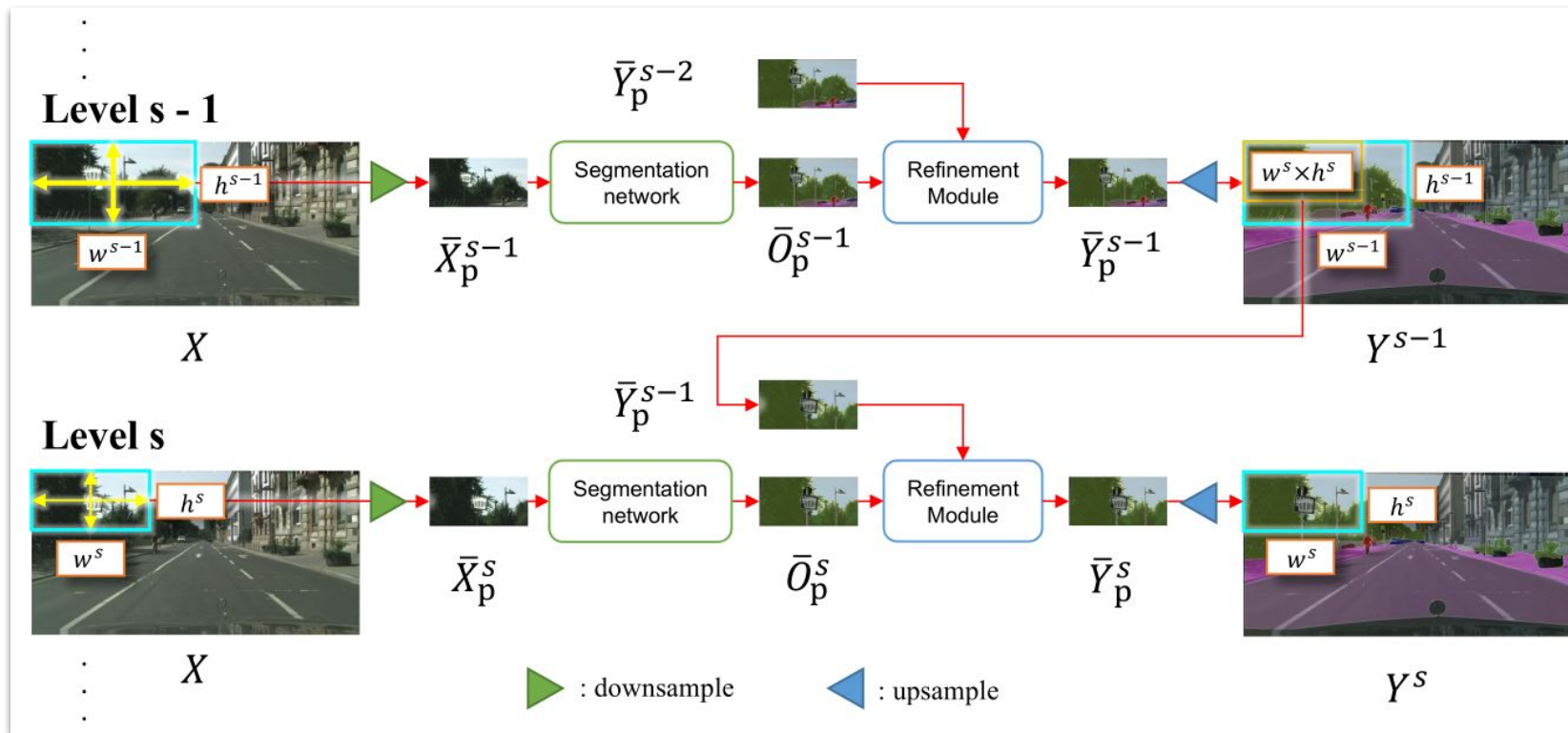
(h) MagNet (Proposed)



MagNet: Multi-stage processing pipeline

Coarse-to-Fine Pipeline

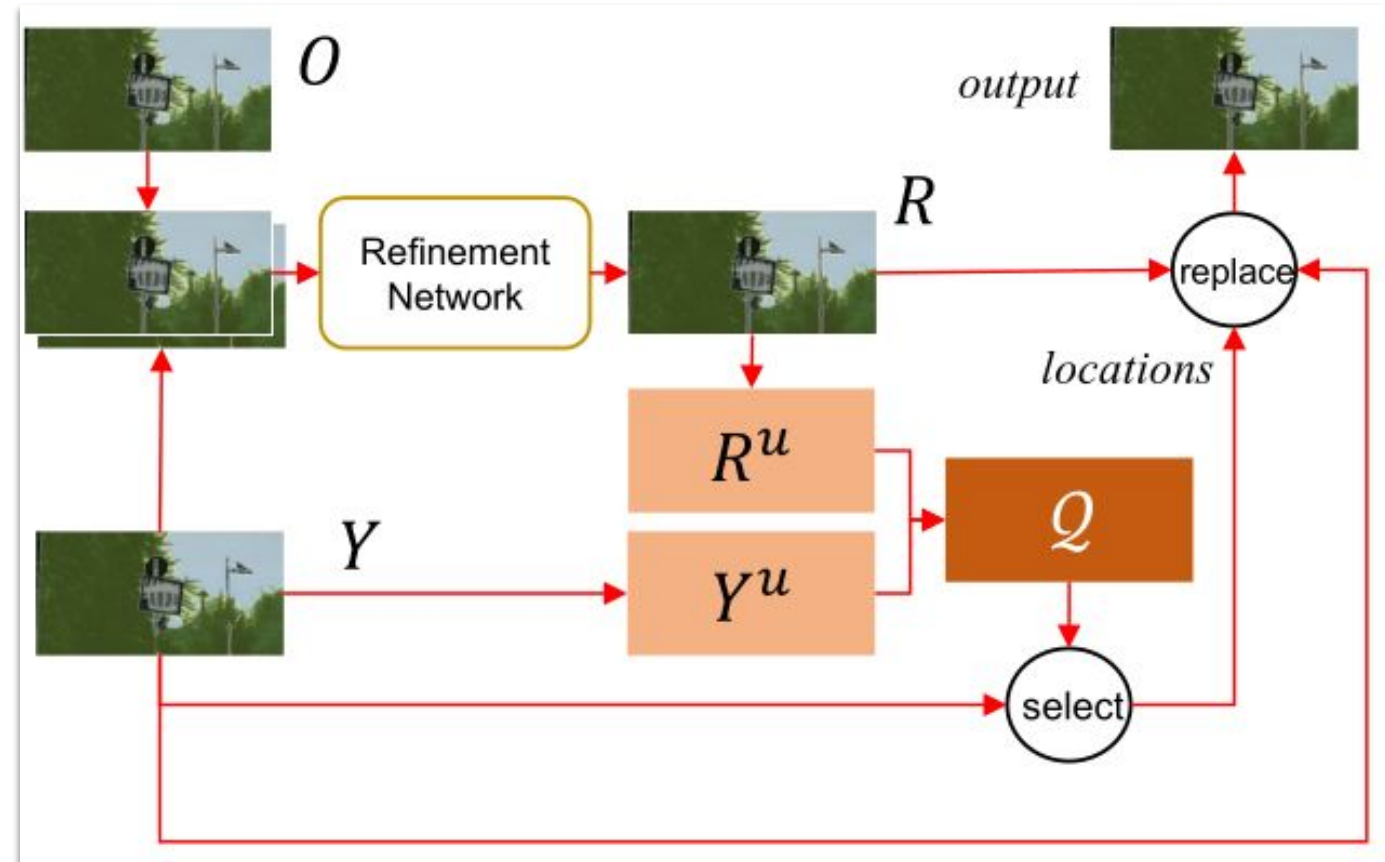
MagNet splits the image into patches and processes them across multiple scales, starting from a coarse global view and moving to finer local details. At each stage, the segmentation output is refined using both the previous cumulative prediction and the new scale-specific prediction. This progressive refinement enables high-resolution segmentation under strict memory constraints.



The Refinement Module

Intelligent Merging

At each scale, MagNet's refinement module merges the coarse prediction from previous stages (Y) with the fine, scale-specific prediction (O). The module first combines them to produce an intermediate map (R), then computes uncertainty maps (Q) for both Y and R .



The Selection Logic

The "Secret Sauce"

The module doesn't replace all pixels. It calculates a score `Q` to find pixels where:

1. The cumulative map (Y) is **uncertain**.
2. The new combined map (R) is **certain**.

This ensures the model updates only meaningful regions instead of rewriting the entire map.

Selection Score (Q)

This formula identifies the most valuable pixels to update.

$$Q = F (Y^u \odot (1 - R^u))$$

Y^u = Uncertainty of the cumulative map

R^u = New map certainty/confidence

F = Median Filtering Function

(used to smooth Q and avoid selecting isolated noisy pixels)

Experiments: datasets and keypoints

Dataset	Content	Resolution	No. classes
Cityscapes	urban scene	2048x2048	19
DeepGlobe	aerial	20448x2448	6
Gleason	histopathology	5000x5000	4

Metric

mIoU
(mean Intersection over Union)

Evaluations

- multiple scale levels
- comparing segmentation approaches

Goal

To prove that MagNet is more accurate and robust than the State-of-the-Art

Key Result: Intermediate Scales are Essential

Refinement Steps (Scales)	mIoU (%) on Cityscapes	Time(s)
256 (Baseline Downsampling)	63.23	0.03
256→512	65.73	0.19
256 → 1024	65.23	0.61
256 → 2048	65.21	2.24
256→512 → 2048	67.13	2.38
256 → 1024→2048	66.95	2.79
256→512→1024→2048	67.57	2.93

Visualizing the Progression

Finer Details Emerge

As the model moves through the scales (from 256x128 up to 2048x1024), the segmentation map becomes progressively more accurate.



MagNet-Fast: The Practical Solution

Full MagNet (Accuracy)

Processes **all** patches at **all** scales. This is computationally expensive.

Example (Cityscapes):

$1 + 4 + 16 + 64 = 85$ patches

Result: Max accuracy (67.57%) but slow (2.93s).

MagNet-Fast (Efficiency)

Processes only a **subset** of patches with the **highest uncertainty** at fewer scales.

Example (Cityscapes):

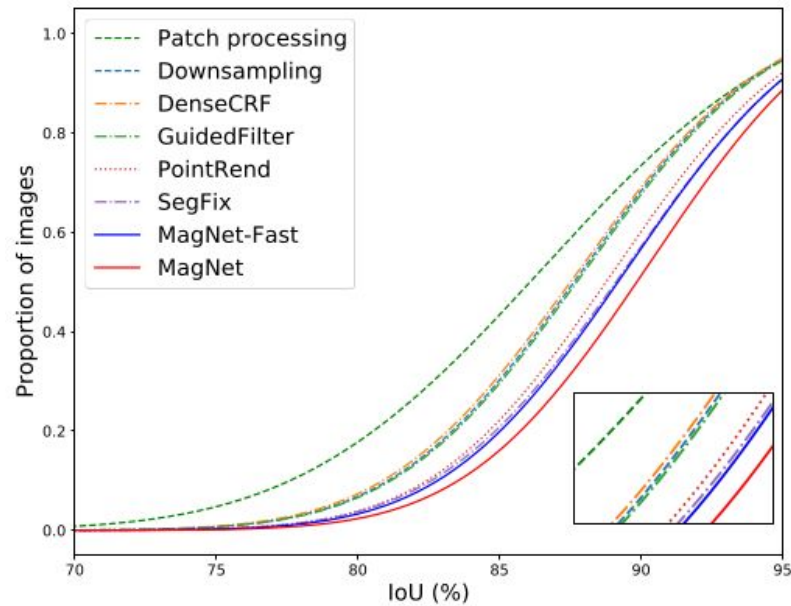
$1 + 3 + 3 = 7$ patches

Result: Huge speed-up (0.32s) & still beats previous SOTA (66.91%).

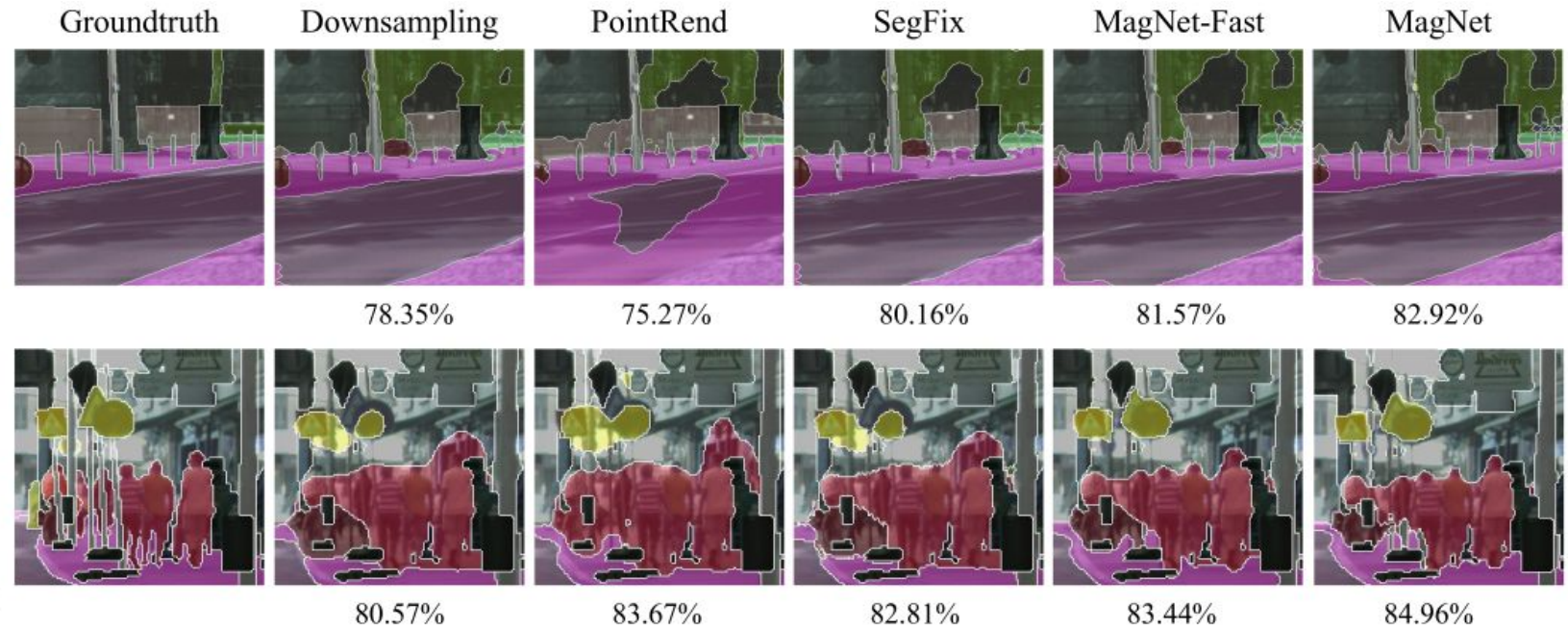
Results: Cityscapes (Urban Scenes)

Model	mIoU (%)	Time (s)
Patching	52.19	1.77
Downsampling	63.23	0.02
PointRend	64.39	0.14
SegFix (Previous SOTA)	65.83	0.38
MagNet-Fast (Ours)	66.91	0.32
MagNet (Ours)	67.57	2.93

Visual Analysis



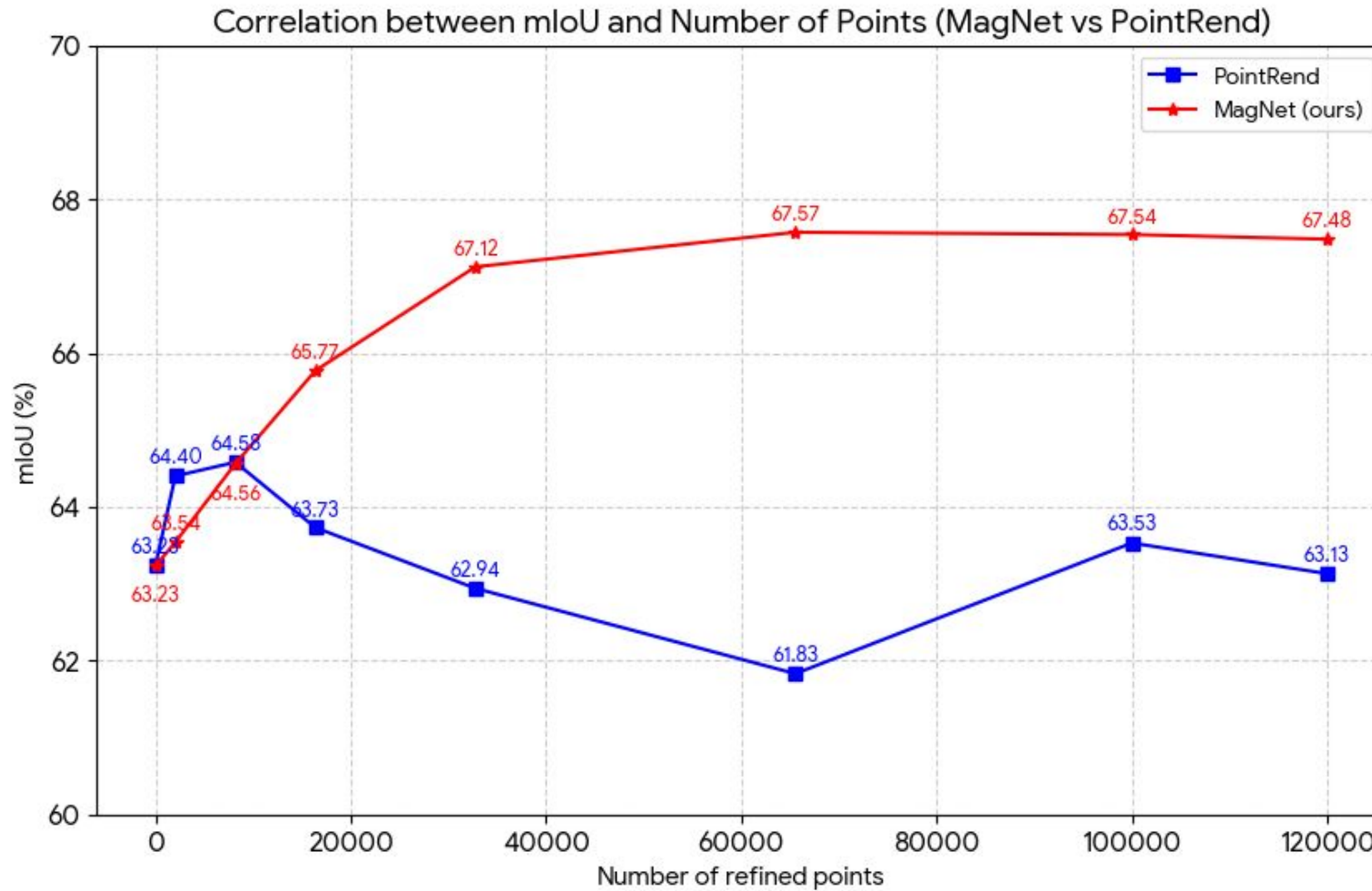
(a) The IoU cumulative distribution



(b) Visualization of refinement methods

MagNet and MagNet-Fast outperform other refinement frameworks on the Cityscapes dataset.

Ablation: Point Selection



MagNet's performance (mIoU) steadily increases and stabilizes as more points are refined. (PointRend's, not shown, becomes unstable and decreases).

Ablation: Model Selection

Model	Backbone	mIoU (%)
Downsampling	DeepLabV3+	52.01
Patch processing	DeepLabV3+	59.64
MagNet	DeepLabV3+	61.99
Downsampling	HRNetV2-W48 + OCR	63.92
Patch processing	HRNetV2-W48 + OCR	54.30
MagNet	HRNetV2-W48 + OCR	68.90

Conclusion



Conclusion

MagNet provides a robust, multi-scale framework for high-resolution segmentation.



Limitations

The full MagNet pipeline is significantly slower (2.93s) than alternatives, creating a speed vs. accuracy trade-off.



Take-Home Message

Progressively refining from coarse-to-fine is a highly effective strategy.